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INTEGRATED SHIPBUILDING QUALITY ENGINEERING

Structural Fabrication, Welding Distortion, Dimensional Accuracy, NDT Interpretation and Defect Prevention

1. INTRODUCTION

Mazagon Dock Shipbuilders Limited (MDL), known as "Shipbuilders to the Nation," is one of India's premier defense sector shipyards under the Ministry of Defence. Operating from Mumbai, MDL specializes in constructing warships, submarines, offshore platforms, and commercial vessels. The shipyard has a distinguished legacy of manufacturing complex naval platforms, requiring extreme precision and adherence to strict defense specifications.

In naval shipbuilding, structural fabrication quality and dimensional accuracy are paramount. Warships and submarines operate under extreme dynamic loads, cyclic hydrostatic pressures, and harsh marine environments. Ensuring structural integrity and dimensional precision directly impacts:

- **Structural Alignment & Stability:** Precise hull geometry ensures hydrodynamic efficiency, propulsion shafting alignment, and proper weight distribution.
- **Fatigue Life & Marine Safety:** High weld quality prevents localized stress concentrations that lead to catastrophic fatigue failure under wave action or underwater shock.
- **Modular Assembly Fit-Up:** Modern shipbuilding uses block assembly methods. Out-of-tolerance blocks cause severe assembly delays, re-work costs, and joint mismatch during final integration.

2. STRUCTURAL FABRICATION QUALITY

Achieving superior fabrication quality requires rigorous technical control across several core mechanical engineering pillars:

- **Dimensional Accuracy:** Verification of overall block dimensions, panel flatness, and frame spacing against approved construction drawings to ensure seamless integration.
- **Weld Quality:** Sound weld deposits free from harmful discontinuities, adhering to specified strength, ductility, and toughness requirements.
- **Fabrication Sequence:** Planned assembly and welding progression designed to balance heat input, minimize residual stress build-up, and control overall distortion.
- **Joint Fit-Up & Structural Alignment:** Control of root gap, bevel angle, root face, and edge alignment (hi-lo) prior to welding, preventing lack of penetration or excessive melt-through.
- **Inspection & Testing:** Non-destructive evaluation (NDE) and stage-wise dimensional verifications at defined quality hold-points.
- **Defect Prevention & Documentation:** Proactive quality controls, qualified procedures (WPS/PQR), welder qualification, and complete material traceability records.

3. ENGINEERING PROBLEM STATEMENT (HYPOTHETICAL CASE STUDY)

> **Educational Scenario Notice**
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> This case study is a **hypothetical educational exercise** designed to illustrate quality engineering principles in shipbuilding. It does not represent an actual incident, internal inspection record, or operational failure at Mazagon Dock Shipbuilders Limited.

1. What Was Observed: During the stage-wise fabrication of a welded structural sub-assembly intended for a naval deck module, dimensional inspection revealed a localized transverse bow distortion exceeding the drawing limit by 4.5 mm over

a 3-meter span. Simultaneously, Visual Testing (VT) identified a surface line indication along the toe of a major load-bearing fillet weld.

2. Investigation Necessity: The deviation requires thorough investigation because assembly fit-up with adjacent hull blocks could be compromised, and the weld toe indication might represent a structural defect that undermines fatigue resistance.

3. Available vs. Required Evidence: Currently available data includes initial stage-wise dimensional reports and a preliminary VT log. Required evidence includes calibrated multi-point laser measurements, review of the approved Welding Procedure Specification (WPS), fit-up records, and secondary Non-Destructive Testing (NDT) reports.

4. Associated Risks: Ignoring the deviation could lead to forced assembly under mechanical restraint (introducing severe secondary stresses), hull block misalignment, or structural failure under service loads.

5. Verification Requirement: The Quality Engineer must verify the governing structural drawing revision, specified engineering tolerances, and applicable NDT acceptance criteria before making any disposition or declaring a non-conformance.

4. WELDING DISTORTION AND DIMENSIONAL ACCURACY

Welding distortion is the permanent, non-uniform dimensional change caused by localized heating and rapid cooling during the fusion welding process.

- **Thermal Mechanism:** High temperature at the weld pool causes localized thermal expansion. The cooler surrounding base metal restrains this expansion, causing compressive yield and plastic deformation in the heated zone. Upon cooling, the weld metal contracts, leaving high tensile residual stresses and permanent distortion.
- **Primary Distortion Modes:**
  - *Angular Distortion:* Transverse shrinkage across asymmetric weld bevels causes rotation around the weld axis.
  - *Longitudinal Shrinkage:* Contraction along the line of the weld seam, leading to shortening or bowing.
  - *Transverse Shrinkage:* Contraction perpendicular to the weld line, causing panel narrowing.
- **Distortion Control Measures:**
  - **WPS & Heat Input Control:** Optimizing welding parameters (voltage, current, travel speed) to minimize heat input ($HI = (V \times I \times 60) / (S \times 1000)$ in kJ/mm).
  - **Balanced Welding Sequence:** Utilizing back-step welding, symmetrical welding around the neutral axis, and balanced pass deposition.
  - **Fit-Up & Alignment:** Maintaining uniform root gaps and bevel prep; using temporary strongbacks or stiffening fixtures where permitted by procedures.
  - **Stage Inspections:** Performing dimensional checks prior to, during, and immediately after welding to catch thermal movement early.

5. WELD DISCONTINUITIES AND NDT METHODS

In welding quality engineering, a **discontinuity** is any interruption in the normal physical structure or property of a weldment (e.g., pore, inclusion, undercut). A discontinuity becomes an **unacceptable defect** only when its size, type, or distribution exceeds the limits specified in the applicable acceptance standard (e.g., ISO 5817 Level B/C).

Discontinuity Type	Physical Description	Primary NDT Method	Criticality in Shipbuilding
Crack (Toe / Hydrogen)	Sharp, planar fracture resulting from high stress, restraint, or hydrogen embrittlement.	VT, PT, MT, UT	**Rejectable.** High stress concentration; initiates premature fatigue failure.
Lack of Fusion (LOF)	Lack of bond between weld metal and base metal or adjacent passes.	UT, RT	**Severe.** Planar internal discontinuity that reduces effective load-bearing area.
Incomplete Penetration	Failure of weld metal to extend completely into the joint root.	UT, RT, VT (open root)	**Severe.** Acts as a sharp notch at the root, lowering fatigue life under cyclic loads.

Discontinuity Type	Physical Description	Primary NDT Method	Criticality in Shipbuilding
Porosity (Gas Pockets)	Spherical or elongated voids caused by trapped gas during solidification.	RT, UT, VT (surface)	**Variable.** Evaluated by volumetric density limits per standard. Non-planar.
Undercut	Groove melted into base metal at weld toe/ root and left unfilled.	VT, Profile Gauge	**Moderate-High.** Reduces plate section thickness and creates stress concentration.

Overview of NDT Methods

- **Visual Testing (VT):** Primary method for surface profile, alignment, undercut, and surface cracks. *Limitation:* Surface-exposed features only.
- **Dye Penetrant Testing (PT):** Uses capillary action to reveal open-to-surface discontinuities on non-porous materials. *Limitation:* Cannot detect sub-surface defects; sensitive to surface cleanliness.
- **Magnetic Particle Testing (MT):** Uses magnetic flux leakage to detect surface and near-surface defects in ferromagnetic materials. *Limitation:* Ferromagnetic metals only; requires demagnetization.
- **Ultrasonic Testing (UT):** Uses high-frequency sound waves to detect and depth-size internal volumetric and planar discontinuities. *Limitation:* Operator skill-dependent; requires couplant and smooth surface.
- **Radiographic Testing (RT):** Uses X-rays or gamma rays to produce volumetric shadow images of internal discontinuities. *Limitation:* Radiation safety hazards; less sensitive to planar laminar defects perpendicular to beam.

6. DIMENSIONAL INSPECTION AND MEASUREMENT RELIABILITY

Before declaring a non-conformance, the engineer must differentiate between actual structural variation and measurement errors:

- **Instrument Suitability & Calibration:** Ensure measuring instruments (laser trackers, total stations, calibrated steel tapes) are appropriate for the required tolerance class and possess valid, traceable calibration certificates.
- **Measurement Repeatability & Reference Datum:** Measurements must be referenced to established shipyard reference lines (baseline, centerline, frame lines). Repeated measurements by independent inspectors confirm data repeatability.
- **Environmental Factors:** Thermal expansion of large steel structures during direct sunlight exposure can introduce transient dimensional changes ($\Delta L = \alpha \cdot L \cdot \Delta T$). Dimensional checks should be performed at stable ambient conditions.

7. QUALITY ENGINEERING TOOLS AND THEIR APPLICATION

Quality tools structure the evidence-based investigation process:

A. Check Sheet: Structured forms used to capture real-time shop-floor observations, such as fit-up gaps, weld pass counts, ambient temperatures, and dimensional measurements across defined grid points.

B. Root Cause Analysis (RCA): A systematic problem-solving process aimed at identifying the fundamental underlying cause of a defect rather than treating superficial symptoms.

C. 5 Whys Analysis (Illustrative Example):

> **Illustrative 5 Whys Exercise (Hypothetical)**
>
> 1. *Why did the deck sub-assembly distort?* → Excessive transverse shrinkage occurred during welding.
> 2. *Why was transverse shrinkage excessive?* → Heat input per pass exceeded the WPS envelope.
> 3. *Why was heat input excessive?* → Travel speed was lower than specified due to heavy weave technique.
> 4. *Why did the welder weave excessively?* → Root fit-up gap was oversized (6 mm instead of 3 mm nominal).
> 5. *Why was the fit-up gap oversized?* → Edge preparation cutting dimensions were not verified prior to assembly tacking.

D. Fishbone (Ishikawa) Diagram Categories:

- **Man:** Welder qualification, technique, training in distortion mitigation.
- **Machine:** Calibration of welding power sources, condition of cutting torches/beveling tools.

- **Method:** WPS parameters, assembly sequence, tack welding details, restraint usage.
- **Material:** Base plate thickness variations, edge bevel geometry, filler metal batch consistency.
- **Measurement:** Calibration of measuring tapes, laser tracker accuracy, datum line reference points.
- **Environment:** Ambient temperature fluctuations, draft/wind conditions during gas shielding.

E. Failure Mode and Effects Analysis (FMEA): Evaluates potential failure modes (e.g., weld toe cracking, block distortion) by assessing Severity (S), Occurrence (O), and Detection (D) to calculate Risk Priority Numbers ($RPN = S \times O \times D$), helping prioritize preventive controls.

F. Corrective Action & Verification of Effectiveness: Corrective actions must address root causes (e.g., revising fit-up inspection hold points). Effectiveness is verified through follow-up audits, trend analysis of subsequent block dimensions, and zero defect recurrence in subsequent fabrications.

8. ROOT CAUSE INVESTIGATION WORKFLOW

A disciplined quality investigation evaluates multiple potential contributors without bias:

1. **Drawing & Revision Control:** Confirm latest revision is in use on the shop floor.
2. **Fit-Up & Joint Prep Inspection:** Verify root opening, bevel angle, and alignment records prior to welding.
3. **WPS & Parameter Compliance:** Audit actual voltage, current, shielding gas flow, and travel speed against WPS limits.
4. **Thermal Input & Sequence:** Check if back-step or balanced welding sequence was followed as per approved shop instructions.
5. **Equipment Calibration:** Verify current/voltage meters on welding power sources.
6. **NDT Indication Re-assessment:** Perform supplementary MT or UT to confirm if the surface VT indication is a crack or cosmetic surface lap.

9. PROPOSED ENGINEERING SOLUTION & EXECUTION PLAN

Step 1: Document & Quarantine
Log the dimensional deviation and NDT indication in an official Non-Conformance Report (NCR). Quarantined structure to prevent unauthorized processing.

Step 2: Drawing & Specification Audit
Verify governing engineering drawing, allowable tolerance limits (±2.0 mm), and NDT acceptance standards.

Step 3: Dimensional Re-measurement
Re-measure using calibrated laser tracker equipment from verified structural datums to rule out measurement error.

Step 4: Advanced NDT Characterization
Perform Magnetic Particle Testing (MT) and Shear Wave Ultrasonic Testing (UT) to determine indication depth, length, and nature (planar vs non-planar).

Step 5: Root Cause Analysis
Convene RCA team (Quality, Welding, Production) using Fishbone and 5 Whys to pinpoint root cause without assumptions.

Step 6: Authorised Disposition & Repair Procedure
If indication exceeds acceptance limits, perform authorized gouging/grinding and re-welding per approved Repair WPS. If distortion is confirmed out of spec, apply approved flame/mechanical straightening procedures.

Step 7: Final Re-inspection & Closeout

Perform 100% VT + MT/UT on repaired region and complete final dimensional check. Document results and close NCR upon Quality Engineering approval.

10. VALIDATION AND EXPECTED IMPACT

Implementing this structured quality engineering approach yields key operational benefits:

- **Enhanced Fit-Up Precision:** Eliminates forced assembly at integration docks, reducing residual assembly stresses.
- **Rigorous Defect Characterization:** Prevents premature grinding or unnecessary weld repairs on acceptable minor indications.
- **Process Stability:** Corrective actions on fit-up tolerances prevent distortion recurrence across identical structural blocks.
- **Complete Traceability:** Maintains complete, auditable quality dossiers compliant with defense vessel classification society requirements.

11. ENGINEERING LEARNING AND KEY TAKEAWAYS

1. Verify First: Always confirm measurements using calibrated instruments and verified datums before declaring a dimensional failure.
2. Indication ≠ Defect: Never label an NDT indication as a defect until evaluated against governing acceptance criteria.
3. Heat Input Control: Distortion control relies on strict WPS parameter adherence and balanced welding sequences.
4. Evidence-Driven RCA: Avoid assumptions or operator blame; use structured tools (5 Whys, Fishbone) backed by shop-floor evidence.
5. Authorised Repairs: All repairs and corrective actions must follow approved procedures, qualified WPS, and formal re-inspection.

12. TECHNICAL INTERVIEW PREPARATION (30 EDUCATIONAL PRACTICE QUESTIONS)

SECTION A: BASIC LEVEL (Questions 1–10)

Q1: What is the fundamental difference between a weld discontinuity and a weld defect?

A weld discontinuity is any interruption in the normal physical structure or property of a weldment (e.g., porosity, slag inclusion). A defect is a specific discontinuity whose size, shape, orientation, or location fails to meet the applicable acceptance criteria of the governing standard, rendering the weld unacceptable.

Q2: What causes angular distortion in fillet and groove welds?

Angular distortion is caused by non-uniform transverse thermal shrinkage across the plate thickness during welding. In single-sided bevels or fillet welds, the wider top portion of the weld deposit contracts more than the root, causing the joint plates to rotate around the weld axis.

Q3: What is heat input in fusion welding, and how is it calculated?

Heat input is the electrical energy supplied by the welding arc per unit length of weld seam. It is calculated as $HI = (V \times I \times 60) / (S \times 1000)$, expressed in kJ/mm, where V is Arc Voltage (volts), I is Welding Current (amperes), and S is Travel Speed (mm/min). Higher heat input increases the thermal zone and thermal distortion.

Q4: Name the five basic NDT methods commonly used in structural fabrication.

Visual Testing (VT), Dye Penetrant Testing (PT), Magnetic Particle Testing (MT), Ultrasonic Testing (UT), and Radiographic Testing (RT).

Q5: What is the main limitation of Liquid Penetrant Testing (PT)?

PT can only detect discontinuities that are open to the surface. It cannot detect subsurface defects or surface defects that are smeared over or sealed by paint, oil, scale, or mechanical peening.

Q6: Why is calibration of measuring instruments critical in dimensional quality control?

Calibration establishes the accuracy and traceability of measuring tools against recognized standards. Uncalibrated or out-of-tolerance instruments can result in false acceptance of defective dimensions or false rejection of compliant assemblies.

Q7: What is the primary purpose of a Welding Procedure Specification (WPS)?

A WPS is a formally qualified written document that provides direction to welders for making production welds compliant with code requirements. It specifies essential variables such as base metal type, filler wire, shielding gas, current/voltage ranges, travel speed, preheat, and interpass temperatures.

Q8: Describe the basic principle of Magnetic Particle Testing (MT).

MT involves magnetizing a ferromagnetic material and applying fine iron particles to the surface. Discontinuities at or near the surface disrupt the magnetic lines of force, creating a magnetic flux leakage field that attracts the particles, forming a visible indication.

Q9: What is the function of a Check Sheet in quality engineering?

A Check Sheet is a structured quality tool used to collect and record data systematically in real time at the shop floor. It facilitates easy data organization, trend identification, and subsequent statistical or graphical analysis.

Q10: What does the term "fit-up" mean in structural welding fabrication?

Fit-up refers to the alignment and positioning of structural components prior to welding. Key parameters include root gap, root face, bevel angle, and plate edge offset (hi-lo alignment). Proper fit-up is essential for achieving full penetration and controlling distortion.

SECTION B: MODERATE LEVEL (Questions 11–20)

Q11: How does a back-step welding sequence help reduce longitudinal bowing distortion?

In back-step welding, short weld increments are deposited in the direction opposite to the overall progression of the weld seam. This technique breaks up continuous longitudinal thermal expansion, distributes thermal stresses more evenly along the joint, and significantly reduces cumulative longitudinal distortion.

Q12: When evaluating an NDT indication, why is it necessary to consult the applicable drawing and revision?

Engineering specifications and acceptance criteria vary depending on the structural criticality of the component. The drawing and revision level specify the exact governing code, quality level (e.g., ISO 5817 Level B or C), and fatigue classification required for that specific joint.

Q13: Differentiate between volumetric NDT methods and planar surface NDT methods.

Volumetric methods (Radiography and Ultrasonics) evaluate the internal volume of a weld, detecting internal voids, inclusions, or sub-surface cracks. Surface/planar methods (VT, PT, MT) focus on surface or near-surface discontinuities, which are often the most critical fatigue crack initiation sites.

Q14: Explain the risk of excessive mechanical restraint during welding distortion control.

While rigid fixtures or restraints prevent physical distortion during welding, they restrict thermal expansion and contraction. This restriction elevates tensile residual stresses within the weldment, which can lead to hot cracking, cold cracking (hydrogen-assisted cracking), or reduced fatigue life in high-strength steels.

Q15: How can ambient thermal expansion affect laser tracker measurements on large ship hull blocks?

Large steel structures undergo measurable thermal expansion under solar heating or ambient temperature swings ($\Delta L = \alpha \cdot L \cdot \Delta T$). If measurements taken at different times of day are compared without thermal compensation, dimensional variations caused purely by temperature will be misidentified as fabrication distortion.

Q16: What role does the "5 Whys" methodology play during a shipbuilding RCA?

The 5 Whys technique drill down through symptomatic problems to uncover the underlying root cause. By repeatedly asking "Why," it moves past immediate physical causes (e.g., welder error) to expose process, procedural, or management system deficiencies (e.g., lack of fit-up verification protocols).

Q17: Why is Ultrasonic Testing (UT) generally preferred over Radiographic Testing (RT) for thick T-joint fillet welds?

T-joint geometry presents severe radiographic path-length variations and scattering, making RT image interpretation difficult. Angle-beam UT can readily direct sound energy through the web and flange to inspect the root zone and weld legs for planar defects like lack of fusion.

Q18: What is the purpose of calculating a Risk Priority Number (RPN) in FMEA?

RPN is calculated as Severity (S) × Occurrence (O) × Detection (D). It provides a quantitative relative score to rank and prioritize potential failure modes, guiding engineering effort and quality resources toward high-risk processes that require immediate preventive controls.

Q19: Explain the difference between corrective action and immediate disposition (correction).

Immediate disposition (correction) fixes an existing non-conforming condition (e.g., grinding out a weld defect and re-welding). Corrective action addresses the root cause of the non-conformance (e.g., recalibrating fit-up tools or revising WPS instructions) to eliminate the cause and prevent recurrence.

Q20: How does joint preparation geometry (e.g., single-V vs. double-V) affect welding distortion?

A single-V bevel requires a large volume of weld metal deposited entirely on one side, leading to high asymmetric shrinkage and severe angular distortion. A double-V bevel allows balanced welding from both sides around the joint's neutral axis, significantly reducing angular distortion.

SECTION C: ADVANCED LEVEL (Questions 21–30)

Q21: You encounter conflicting dimensional inspection results between a laser tracker and conventional steel tape measurements on a deck module. How do you resolve this dispute?

First, check calibration certificates, temperature compensation settings, and procedure compliance for both instruments. Verify that both methods used identical reference datums and baseline temperature corrections. If discrepancy persists, perform a controlled re-measurement using a validated 3D optical measurement system with environmental monitoring to establish definitive dimensional data.

Q22: A surface Linear Magnetic Particle (MT) indication is detected in the heat-affected zone (HAZ) of a high-strength naval steel weldment. What steps should the Quality Engineer take?

Do not immediately grind or reject. First, record indication length and orientation. Verify if it is a true crack or a non-relevant indication caused by magnetic permeability shifts or geometry. If confirmed as a crack, perform Ultrasonic Testing (UT) to determine depth. Consult the engineer of record and approved repair procedure before authorizing controlled excavation, MT verification of defect removal, and re-welding per qualified WPS.

Q23: How do you differentiate between welding-induced distortion and initial plate mill flatness out-of-tolerance in a fabricated deck panel?

Review stage-wise incoming material inspection records and pre-assembly dimensional logs taken before welding. Plate mill bow or wave exists prior to cutting and welding. Welding distortion manifests near weld seams and increases progressive with weld pass deposition. Comparing pre-weld grid measurements with post-weld laser scans isolates thermal shrinkage from original material curvature.

Q24: Explain how residual stress distribution differs between flame-straightened panels and cold mechanical press-straightened panels.

Flame straightening introduces localized thermal yield that creates counter-balancing tensile residual stresses upon cooling, modifying the overall residual stress pattern. Cold mechanical straightening forces plastic deformation, introducing high compressive and tensile mechanical residual stress fields that may affect fatigue resistance or buckling strength under high dynamic service loads.

Q25: Why is hydrogen-assisted cold cracking (HACC) particularly dangerous in naval structural welds, and how is it prevented?

HACC can occur hours or days after welding completes (delayed cracking) in high-strength steels. It results from hydrogen presence, susceptible microstructure (martensite), and high tensile stress. Prevention requires strict low-hydrogen consumable handling (baking/drying), adequate preheat and interpass temperature maintenance, and post-weld heat soak to allow hydrogen effusion.

Q26: During NDT audit, an inspector reports "lack of fusion" using Straight Beam UT, but RT reports "no defects found." Explain why this occurs.

Lack of fusion is a tight, planar discontinuity. If the radiography X-ray beam is not oriented parallel to the fusion plane, the density change is insufficient to project a visible shadow on film/digital detector. Ultrasonic shear waves directed perpendicular to the fusion plane produce strong reflections, making UT vastly superior for planar LOF detection.

Q27: How can Quality Engineering prevent "recurrent distortion" across a series of 10 identical ship structural blocks?

Implement a closed-loop Quality Control process: Analyze baseline data from Block 1 using laser scanning; perform Root Cause Analysis on identified shrinkage patterns; refine the WPS heat input limits; optimize assembly and welding sequence (e.g., symmetrical dual-welder progression); implement pre-cambering or mechanical restraint where authorized; and monitor statistical process control (SPC) trends across subsequent blocks.

Q28: An engineering specification allows a maximum local angular distortion of 3 mm over 1 m. Measured distortion is 4.2 mm. What engineering considerations dictate whether a concession can be granted?

The engineer must evaluate: 1) Structural criticality of the location (primary load-bearing vs non-structural panel); 2) Impact on fatigue life and buckling resistance under operational hydrodynamic loads; 3) Downstream integration consequences with mating blocks; 4) Feasibility of flame/mechanical alignment without introducing harmful residual stresses. Concession requires formal engineering justification and client/classification society approval.

Q29: How do you verify the effectiveness of a Corrective Action implemented for recurring fillet weld undercut?

Verification requires a multi-stage approach: 1) Audit shop floor compliance with revised WPS (voltage/travel speed limits); 2) Perform 100% VT and optical gauge profiling on the next 20 production joints; 3) Analyze inspection check sheets over a 30-day period using SPC p-charts to confirm defect rate reduction; 4) Ensure zero recurring NCRs prior to formal CAPA closeout.

Q30: Why must non-destructive testing acceptance standards be explicitly linked to structural design criteria rather than applying a universal "zero-defect" rule?

A universal "zero-defect" policy is economically impractical and technically unnecessary. Acceptance criteria (e.g., ISO 5817 Levels B, C, D) align quality requirements with component stress levels and fatigue exposure. Non-critical secondary structures can safely tolerate minor non-planar discontinuities without compromising structural integrity, whereas critical hull joints require stringent defect limits.

13. REFERENCES

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